20251020EB_MN (seahorse, vLH51, vLH103)

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2025年10月20日 星期一

split iPSCs into a full 6-well plate on the Friday before differentiation
differentiate 5 of 6-wells with 10mL T2 in cell-repellent plates
keep some iPSCs

2025年10月22日 星期三

T2 -> T3

2025年10月24日 星期五

T3 -> T4 (~14 mL for over-weekend)

2025年10月27日 星期一

T4 -> T5

2025年10月29日 星期三

T5 -> T6

2025年10月31日 星期五

T6 -> T6 (~14mL for over-weekend)

2025年11月3日 星期一

T6 -> T7
Dissociation

1. Coat 0.01% Poly-L-Ornithine (PLO) Solution with sodium borate (1:100) for 4-5 hrs
2. Remove the PLO and wash the plate with PBS once
3. Coat the plate with laminine for 1hr.
4. Collect EBs in 50mL falcon and spin them down with 200rcf, 1min. Remove the supernatent (can leave some liquid to avoid remove EBs).
5. Add trypsin (usually 3-4 mL/10-cm plate) and pipette it up and down with 5mL serological pipette.
6. Put the tube into 37C water bath. Pipette the cells with P1000 every 5 min round 8-10 times until 15 min. (3 times in total. There might be more resistance at the 2nd round)

7. Add 2x volumn of Heat Inactivated FBS into the tube and pipette it with 5mL serological pipette gently. (Try to make the viscous cluster get through the tip to break it down)
8. Add 2x volumn of CTWM and pipette it with 5mL serological pipette gently.
9. Filter the cell with 40um filter into a new 50mL tube
10. Spin the cell down at 300rcf, 7min. Remove the supernatent.
11. Add T7 (~5mL) to resuspend the cells and count it.
12. Plate the cells with T7. (350-500K/well for 12well plate; 1-1.5M/well for 6-well plate)

cell info

	A	B
1	cell	passage
2	WT11	

Table1

	A	B
1	run1	volume (mL)
2	trypsin	3
3	HI-FBS/CTWM	6
4	run2	*take some undissolved EB
5	trypsin	2
6	HI-FBS/CTWM	4

Table3

	A	B	C	D	E	F	G	H	I
1	conc. (M/mL)	survival	total volume	tota cell count	plate#	well plate	cell count/well (K)	number of wells	volume
2	4.26	99	7	29.82	A	96	60	36	507
3					B	24	250	16	939
4					C,D	6	1972	12	the rest

2025年11月4日 星期二

Change T7 -> T7 if added virus

2025年11月5日 星期三

T7-> T7 if not added virus

2025年11月6日 星期四

T7 -> T8

2025年11月7日 星期五

1/2 T8 -> 1/2 T8

2025年11月10日 星期一

1/2 T8 -> 1/2 T8

2025年11月11日 星期二

Treat cells for seahorse at 13:45 for 24h

2025年11月12日 星期三

1/2 T8 -> 1/2 T8

Seahorse Plate A

condition: 1.5uM oligomycin, 1uM FCCP, 0.5uM Rot/AA

1. Rehydrate the cartridge the day before.
2. Turn on the seahorse machine.
3. Prepare drugs from stock as the drug table below with two glucose concentrations.
4. every well wash with 90-100uL media with corresponding glucose concentration
5. add 175uL of media to the cells and add 2.5mM glucose media to all the blank wells.
6. incubate the plate for 1 hour to deplete CO2.
7. add drug to the cartridge as below. Add H2O for all the remaining wells.

drug in cartridge layout

	A	B
1	A (oligo)	B (FCCP)
2	C (Rot/AA)	D (H2O)

drug calculation per glucose concentration

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1		stock (uM)	1st dilution factor	total volume for 1st dil.	stock volume	media volume for 1st dil.	conc. after 1st dilution	working (uM)	loading volume	well volume before loaded	loading dilution	loading conc.(uM)	dilution factor from 1st dil.	well# for each condition	total loading volume	volume of 1st dil.	volume of media
2	oligo	10000	10	20	2	18	1000	1.5	25	175	8	12	83.3333333333	14	400	4.8	395.2
3	FCCP	10000	20	20	1	19	500	1	25	200	9	9	55.5555555556	14	400	7.2	392.8
4	Rot/AA	10000	10	10	1	9	1000	0.5	25	225	10	5	200	14	400	2.0	396.0
5	2DG	2000000					2000000	50000	25	250	11	550000	3.6363636364	18	500	137.5	225.0
6	medium								275					14	4400		4400
7						46								12240			5584
8														17870			5630

use 2mM glutamine for this time (instead of 4mM)

prepare seahorse basal media

	A	B	C	D	E	F	G
1		stock (mM)	**working (mM)**	loading dilution	total loading volume	volume of media	
2	glucose	1100	**2.5**	440	18500	42.0	18273.0
3		1100	0.15	7333.3333333333	6000	0.8	5939.2
4	glutamine	200	**2**	100	18500	185.0	
5		200	**2**	100	6000	60.0	
6							
7							

Well2

	1	2	3	4	5	6	7	8	9	10	11	12
A												
B		cells don't look well	cells look not optimal									
C			cell died									
D									<- 2.5mM gluc 24h			
E		cells don't look well							<- 0.15mM gluc 24h			
F												
G												
H												

seahorse treatment

	1	2	3	4	5	6	7	8	9	10	11	12
A												
B		1	1	1	2	2	1	1				
C		2	2	2	2	2	1	1				
D		3	3	3	3	3	3	3	<- 2.5mM gluc 24h			
E		4	4	4	4	4	4	4	<- 0.15mM gluc 24h			
F												
G												
H												

Cell counting (normalization) using total protein concentration

2025年11月14日 星期五

Imaging GEM (PlateB)

100x new spinning disk confocal: 0.11um/px

*note:

- A5 is really hard to find cells
- vLH103 this time is great! Cells express great density of GEMs.
- vLH51: ROI=600x900; vLH103: ROI=704x704

Well1

	1	2	3	4	5	6
A					2-2	2-2
B					2-1	2-1
C					1	1
D						

Table4

	A	B	C	D	E	F	G	H	I	J
1			vLH51				vLH103			
2	condition#	condition	treat time	image time	soma	neurite	treat time	image time	total position	
3	1	2.5mM glucose 1h	13:00	14:00	1-15	1-10	14:40	15:56	20	
4	2-1	0.15mM glucose 1h	13:40	14:40	16-30	11-20	15:03	16:13	20	
5	2-2	0.15mM glucose 1h	14:15	15:10	31-45	21-30	15:40	16:40	12	A6 condition is less optimal

Treat cells with vLH103 with SirDNA

3*400/5000=0.24

dilute 1:20 of SirDNA -> 0. in 4.8 uL 0.15mM media --> 1.6 uL/well

103_O.C. , 100X, 1xzoom, 600x900, 12-bit

	A	B	C	D	E
1	channel	bit	power (%)	exposure time	note
2	488	12	100	10ms	
3	640	12	10	100 ms	snapshot
4	488	12	35	50ms	

vLH51_fileid-conditions

	A	B	C	D
1	image	treatment	treat_time	type
2	soma_vLH51_MN_100x_1xzoom_001	2.5mM gluc	1h	movie
3	soma_vLH51_MN_100x_1xzoom_002	2.5mM gluc	1h	movie
4	soma_vLH51_MN_100x_1xzoom_003	2.5mM gluc	1h	movie
5	soma_vLH51_MN_100x_1xzoom_004	2.5mM gluc	1h	movie
6	soma_vLH51_MN_100x_1xzoom_005	2.5mM gluc	1h	movie
7	soma_vLH51_MN_100x_1xzoom_006	2.5mM gluc	1h	movie
8	soma_vLH51_MN_100x_1xzoom_007	2.5mM gluc	1h	movie
9	soma_vLH51_MN_100x_1xzoom_008	2.5mM gluc	1h	movie
10	soma_vLH51_MN_100x_1xzoom_009	2.5mM gluc	1h	movie
11	soma_vLH51_MN_100x_1xzoom_010	2.5mM gluc	1h	movie
12	soma_vLH51_MN_100x_1xzoom_011	2.5mM gluc	1h	movie
13	soma_vLH51_MN_100x_1xzoom_012	2.5mM gluc	1h	movie
14	soma_vLH51_MN_100x_1xzoom_013	2.5mM gluc	1h	movie
15	soma_vLH51_MN_100x_1xzoom_014	2.5mM gluc	1h	movie
16	soma_vLH51_MN_100x_1xzoom_015	2.5mM gluc	1h	movie
17	soma_vLH51_MN_100x_1xzoom_016	0.15mM gluc	1h	movie
18	soma_vLH51_MN_100x_1xzoom_017	0.15mM gluc	1h	movie
19	soma_vLH51_MN_100x_1xzoom_018	0.15mM gluc	1h	movie
20	soma_vLH51_MN_100x_1xzoom_019	0.15mM gluc	1h	movie
21	soma_vLH51_MN_100x_1xzoom_020	0.15mM gluc	1h	movie
22	soma_vLH51_MN_100x_1xzoom_021	0.15mM gluc	1h	movie
23	soma_vLH51_MN_100x_1xzoom_022	0.15mM gluc	1h	movie
24	soma_vLH51_MN_100x_1xzoom_023	0.15mM gluc	1h	movie
25	soma_vLH51_MN_100x_1xzoom_024	0.15mM gluc	1h	movie
26	soma_vLH51_MN_100x_1xzoom_025	0.15mM gluc	1h	movie
27	soma_vLH51_MN_100x_1xzoom_026	0.15mM gluc	1h	movie
28	soma_vLH51_MN_100x_1xzoom_027	0.15mM gluc	1h	movie
29	soma_vLH51_MN_100x_1xzoom_028	0.15mM gluc	1h	movie
30	soma_vLH51_MN_100x_1xzoom_029	0.15mM gluc	1h	movie
31	soma_vLH51_MN_100x_1xzoom_030	0.15mM gluc	1h	movie
32	soma_vLH51_MN_100x_1xzoom_031	0.15mM gluc	1h	movie
33	soma_vLH51_MN_100x_1xzoom_032	0.15mM gluc	1h	movie

34	soma_vLH51_MN_100x_1xzoom_033	0.15mM gluc	1h	movie
35	soma_vLH51_MN_100x_1xzoom_034	0.15mM gluc	1h	movie
36	soma_vLH51_MN_100x_1xzoom_035	0.15mM gluc	1h	movie
37	soma_vLH51_MN_100x_1xzoom_036	0.15mM gluc	1h	movie
38	soma_vLH51_MN_100x_1xzoom_037	0.15mM gluc	1h	movie
39	soma_vLH51_MN_100x_1xzoom_038	0.15mM gluc	1h	movie
40	soma_vLH51_MN_100x_1xzoom_039	0.15mM gluc	1h	movie
41	soma_vLH51_MN_100x_1xzoom_040	0.15mM gluc	1h	movie
42	soma_vLH51_MN_100x_1xzoom_041	0.15mM gluc	1h	movie
43	soma_vLH51_MN_100x_1xzoom_042	0.15mM gluc	1h	movie
44	soma_vLH51_MN_100x_1xzoom_043	0.15mM gluc	1h	movie
45	soma_vLH51_MN_100x_1xzoom_044	0.15mM gluc	1h	movie
46	soma_vLH51_MN_100x_1xzoom_045	0.15mM gluc	1h	movie
47	neurite_vLH51_MN_100x_1xzoom_001	2.5mM gluc	1h	movie
48	neurite_vLH51_MN_100x_1xzoom_002	2.5mM gluc	1h	movie
49	neurite_vLH51_MN_100x_1xzoom_003	2.5mM gluc	1h	movie
50	neurite_vLH51_MN_100x_1xzoom_004	2.5mM gluc	1h	movie
51	neurite_vLH51_MN_100x_1xzoom_005	2.5mM gluc	1h	movie
52	neurite_vLH51_MN_100x_1xzoom_006	2.5mM gluc	1h	movie
53	neurite_vLH51_MN_100x_1xzoom_007	2.5mM gluc	1h	movie
54	neurite_vLH51_MN_100x_1xzoom_008	2.5mM gluc	1h	movie
55	neurite_vLH51_MN_100x_1xzoom_009	2.5mM gluc	1h	movie
56	neurite_vLH51_MN_100x_1xzoom_010	2.5mM gluc	1h	movie
57	neurite_vLH51_MN_100x_1xzoom_011	0.15mM gluc	1h	movie
58	neurite_vLH51_MN_100x_1xzoom_012	0.15mM gluc	1h	movie
59	neurite_vLH51_MN_100x_1xzoom_013	0.15mM gluc	1h	movie
60	neurite_vLH51_MN_100x_1xzoom_014	0.15mM gluc	1h	movie
61	neurite_vLH51_MN_100x_1xzoom_015	0.15mM gluc	1h	movie
62	neurite_vLH51_MN_100x_1xzoom_016	0.15mM gluc	1h	movie
63	neurite_vLH51_MN_100x_1xzoom_017	0.15mM gluc	1h	movie
64	neurite_vLH51_MN_100x_1xzoom_018	0.15mM gluc	1h	movie
65	neurite_vLH51_MN_100x_1xzoom_019	0.15mM gluc	1h	movie
66	neurite_vLH51_MN_100x_1xzoom_020	0.15mM gluc	1h	movie
67	neurite_vLH51_MN_100x_1xzoom_021	0.15mM gluc	1h	movie
68	neurite_vLH51_MN_100x_1xzoom_022	0.15mM gluc	1h	movie
69	neurite_vLH51_MN_100x_1xzoom_023	0.15mM gluc	1h	movie
70	neurite_vLH51_MN_100x_1xzoom_024	0.15mM gluc	1h	movie

71	neurite_vLH51_MN_100x_1xzoom_025	0.15mM gluc	1h	movie
72	neurite_vLH51_MN_100x_1xzoom_026	0.15mM gluc	1h	movie
73	neurite_vLH51_MN_100x_1xzoom_027	0.15mM gluc	1h	movie
74	neurite_vLH51_MN_100x_1xzoom_028	0.15mM gluc	1h	movie
75	neurite_vLH51_MN_100x_1xzoom_029	0.15mM gluc	1h	movie
76	neurite_vLH51_MN_100x_1xzoom_030	0.15mM gluc	1h	movie

Table2

	A	B	C	D
1		treatment	treat time	type
2	vLH103_MN_100x_1xzoom_0001	2.5mM gluc	1h	movie
3	vLH103_MN_100x_1xzoom_0002	2.5mM gluc	1h	snapshot
4	vLH103_MN_100x_1xzoom_0003	0.15mM gluc	1h	movie
5	vLH103_MN_100x_1xzoom_0004	0.15mM gluc	1h	snapshot
6	vLH103_MN_100x_1xzoom_0005	0.15mM gluc	1h	movie
7	vLH103_MN_100x_1xzoom_0006	0.15mM gluc	1h	snapshot